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SHORT COMMUNICATION

Whole-genome mining and in silico analysis of *FAD* gene family in *Brassica juncea*

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Abstract

Brassica juncea is one of important oilseed crops, and FA compositions determine quality of vegetable oil. Although fatty acid desaturases (FADs) are mainly responsible for modifying seed FA compositions, genome-wide analysis of *FAD* gene family in *B. juncea* (*BjuFAD*) is not reported. Here, we identified 57 *BjuFAD* genes in *B. juncea* genome using homology searches. These *FAD* genes were unevenly distributed in 16 chromosomes and 3 scaffolds. Phylogenetic analysis showed that *BjuFAD* genes were divided into seven subfamilies. Exon–intron organizations, intron patterns and MEME motifs were highly conserved within each of *BjuFAD* subfamilies. Moreover, subcellular locations of deduced BjuFAD proteins and *cis*-acting elements in *BjuFAD* promoters were predicted and analyzed using online software. In addition, 16 SSR loci were totally found in *BjuFAD* genes/promoters. This work provides a basis for further function study of *BjuFAD* genes in quality improvement of *B. juncea* seed oil and in plant development as well as stress response.

Keywords *Brassica juncea* · Fatty acid desaturase · Phylogentic analysis · Gene structure · Conserved motif · *cis*-Acting element

Abbreviations

FA	Fatty acid
FAD	Fatty acid desaturase
MEME	Multiple expectation maximization for motif elicitation
NJ	Neighbour-joining
SSR	Simple sequence repeats

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In high plants, unsaturated FAs are critical components of cell membrane lipids and seed storage lipids (Ohlrogge and Browse 1995). Additionally, some unsaturated FAs, e.g. α-linolenic acid, also act as precursors of phytohormone JA (Weber 2002). The degree of FA unsaturation in membrane lipids is of great importance to maintain the appropriate fluidity and integrity of cell membrane in various environmental factors' response and adaption (Upchurch 2008). Notably, content and composition of unsaturated FAs determine oxidative stability and nutritional value of vegetable oil (Scarth and Tang 2006).

It is well known that FADs are key enzymes for biosynthesis of unsaturated FAs by introducing unsaturated double bonds into fatty acid acyl chains (Ohlrogge and Browse 1995). Previous studies demonstrated that FADs in plants are divided into two classes: soluble and membrane-bound types (Shanklin and Cahoon 1998; Sperling et al.

2003). Soluble FADs are mainly found in plant plastid, e.g., SAD/FAB2 (Sperling et al. 2003). FAB2 is responsible for introducing the first double bond into 18:0-ACP or 16:0-ACP (Shanklin and Somerville 1991). Membrane-bound FADs exist in plant, algae, animal, fungi and bacteria, and contain a wide variety of FAD subfamilies, which are located in plastids or ER (Lou et al. 2014). These FADs catalyze the desaturation of various carbon position of fatty acyl chains, such as Δ3, Δ4, Δ5, Δ6, Δ7, Δ8, Δ9, Δ12, Δ13 and Δ15 (Sperling et al. 2003; Hashimoto et al. 2008). FAB2s are characterized with two conserved D/EXXH histidine motifs (Tocher et al. 1998; Shanklin and Cahoon 1998), whereas membrane-bound FADs generally contain three conserved histidine boxes, $H(X)_{3-4}H$, $H(X)_{2-3}HH$ and $H/Q(X)_{2-3}HH$ (Sperling et al. 2003). These histidine boxes and iron ions possibly constitute the catalytic center of the desaturase (Los and Murata 1998).

Plant FADs function in response to abiotic/biotic stress and also act in phytohormone signal pathway (Nishiuchi and Iba 1998; Upchurch 2008; Xue et al. 2017a, b). For example, transgenic tomato plants overexpressing *ω-3 FAD* gene increased the tolerance to chilling temperatures and salinity stress (Wang et al. 2014; Dominguez et al. 2010), whereas silencing of three *GmFAD3* isoforms enhanced the susceptibility of *Bean pod mottle virus* in soybean (Singh et al. 2011). The expression of both cotton *FAD2-3* and *FAD2-4* genes was induced under cold and light stress (Kargiotidou et al. 2008). Given that the promoter of *Brassica napus FAB2* gene had a putative ABA responsive element (ABRE), ABA induced its transcriptional expression in embryo (Slocombe et al. 1994).

Brassica juncea (AABB, $2n = 36$) is an allopolyploid *Brassica* species formed by natural hybridization between *Brassica rapa* (AA, $2n = 20$) and *Brassica nigra* (BB, $2n = 16$) (Yang et al. 2016). It is mainly cultivated with oil- and vegetable-use subvarieties, such as vegetable mustard, oilseed crops and condiment crops (Chen et al. 2013; Yang et al. 2016). The nutritional and industrial value of *B. juncea* seed oil is mainly determined by seed FA composition (Scarth and Tang 2006). It is well known that FADs play a pivotal role in modifying FA compositions of seed oil. To date, sequenced genomes have facilitated genome-wide analysis of *FAD* genes in numerous plants, e.g., rapeseed and its parental species (Xue et al. 2018), soybean (Chi et al. 2011), flax (You et al. 2014), cotton (Liu et al. 2015) and cucumber (Dong et al. 2016). Although genome of *B. juncea*, one of important oilseed crops, has been deposited, little is reported on systematic identification of *B. juncea FAD* (*BjFAD*) gene family at the whole-genome scale. So far, only few *FAD* genes have been isolated and characterized (Vageeshbabu et al. 1996; Singh et al. 1995; Garg et al. 2000).

Here, based on bioinformatics method (Supplementary Material), 57 *BjuFAD*, 45 *BraFAD* and 46 *BniFAD* genes were identified in total in *B. juncea*, *B. rapa* and *B. nigra* genome, respectively (Supplementary Table S1–S4; Table 1). Genomic sequences of *BjuFAD* genes varied from 855 to 9473 bp in length, and their coding sequences ranged from 651 to 3417 bp. BjuFAD proteins varied from 217 to 1139 aa in length. Theoretical MWs ranged from 24.94 to 129.77 kDa and pI values ranged from 5.25 to 10.26. Among 57 BjuFAD proteins, all BjuFAB2 members contained pI values < 7 and rest of BjuFADs except for BjuADS2 had pI values > 7. In addition, distribution of conserved histidine-boxes in BjuFADs was shown in Supplementary Table S5. BjuFAD3s and BjuFAD2s (except for BjuFAD2.3/4) harbored a ER-located signal "KSKIN" and "YNNKL", respectively.

Phylogenetic analysis indicated that *FAD* gene family from *B. juncea*, *B. rapa*, *B. nigra* and *A. thaliana* was divided into 7 subgroups, i.e. FAB2/SAD, ADS/FAD5, SLD, FAD4, DES1L, ω-3 and ω-6 (Fig. 1). FAB2/SAD subfamily contained genes encoding stearoyl-ACP desaturase, which was soluble FAD only present in plants, and performed the conversion of 18:0-ACP to $18{:}1^{\Delta 9}$-ACP (Shanklin and Somerville 1991). *B. juncea*, *B. rapa*, *B. nigra* and *A. thaliana* contained 12, 7, 8 and 7 *FAB2* genes, respectively. ADS/FAD5 subfamily included acyl-CoA desaturase-like genes. *A. thaliana* ADS family contained 9 genes, exhibiting diverse regiospecificity for catalyzing introduction of double bonds at n-6/7/9 or Δ7/9 carbon position (Smith et al. 2013). *B. juncea* possessed 11 ADS genes, while *B. rapa* and *B. nigra* comprised 20 and 19, respectively, suggesting gene loss in *B. juncea* genome after a hybridization between ancestors of *B. rapa* and *B. nigra* (Yang et al. 2016). SLD subfamily consisted of genes encoding sphingolipid Δ-8 desaturase, which harbored a cytochrome b5-like domain in N-termini similar to front-end FAD (Meesapyodsuk and Qiu 2012). Seven *SLD* genes

Table 1 The total number of *FAD* gene in each subfamily of *B. juncea*, *B. rapa*, *B. nigra* and *A. thaliana*

Category	*B. juncea*	*B. rapa*	*B. nigra*	*A. thaliana*
Total	57	45	46	25
FAB2/SAD	12	7	8	7
SLD	7	4	4	2
DES1L	2	1	1	1
ADS/FAD5	11	20	19	9
FAD4	7	4	4	1
ω-6	8	3	4	2
ω-3	10	6	6	3

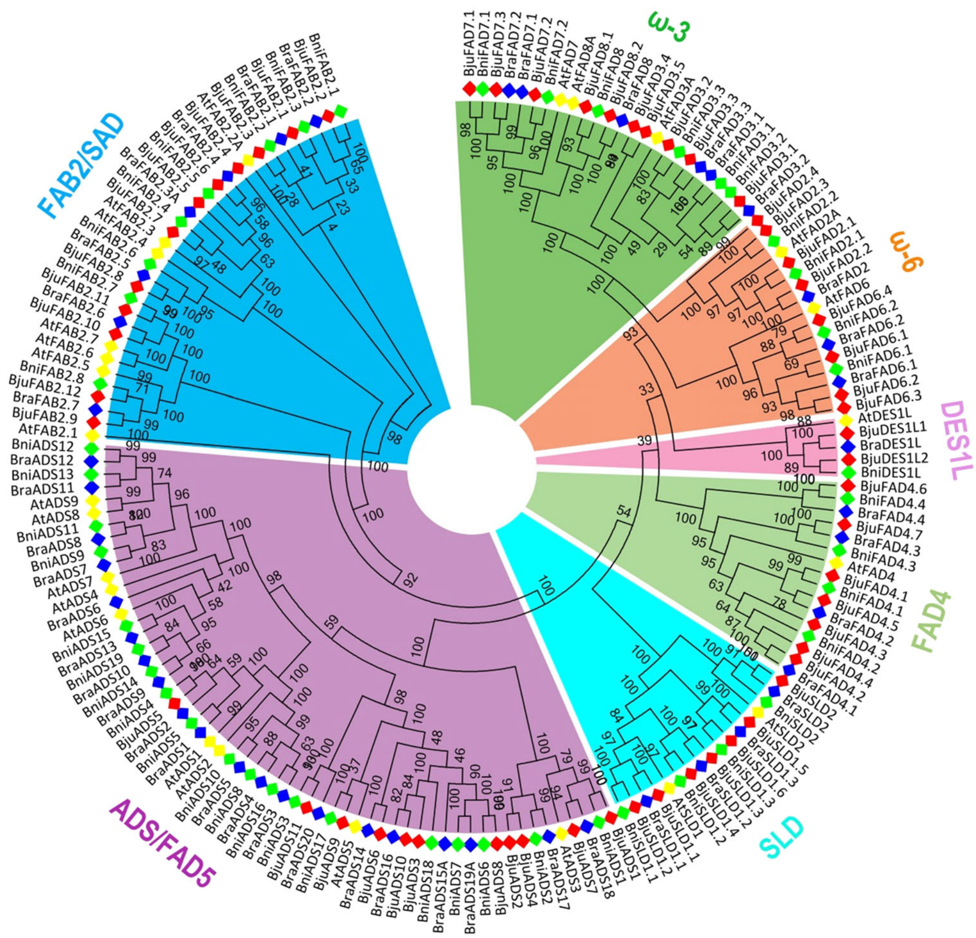

Fig. 1 Phylogenetic relationships of FAD proteins among *B. juncea*, *B. rapa*, *B. nigra* and *A. thaliana*. BjuFADs, BraFADs, BniFADs and AtFADs were indicated with red, blue, green and yellow solid squares, respectively. AtFADs, BraFADs and BniFADs accession numbers were shown in Supplementary Table 1, 3 and 4

were found in *B. juncea*, 4 genes in either *B. rapa* or *B. nigra*, and 2 genes in *A. thaliana*. FAD4 subfamily contained genes encoding Δ3 FAD, catalyzing the desaturation of 16:0 to $16{:}1^{\Delta 3}$ (Gao et al. 2009). 7, 4, 4 and 1 *FAD4* genes were present in *B. juncea*, *B. rapa*, *B. nigra*, and *A. thaliana*, respectively. DES1L subfamily comprised genes encoding sphingolipid Δ-4 FAD (Michaelson et al. 2009). *B. juncea* contained 2 *DES1L* genes, whereas only 1 gene was present in each of *B. rapa*, *B. nigra* and *A. thaliana*. The ω-6 subfamily comprised *FAD2/6* genes encoding ω-6/Δ-12 FAD, which performed the conversion of $16{:}1^{\Delta 9}$ to $16{:}2^{\Delta 9,12}$or $18{:}1^{\Delta 9}$ to$18{:}2^{\Delta 9,12}$ (Ohlrogge and Browse, 1995). Eight *ω-6 FAD* genes were found in *B. juncea*, whereas 3, 4 and 2 genes were present in *B. rapa*, *B. nigra* and *A. thalina*, respectively. The ω-3 subfamily included *FAD3/7/8* genes encoding ω-3/Δ-15 FAD, which catalyzed the desaturation of $16{:}2^{\Delta 9,12}$ to $16{:}3\ ^{\Delta 9,12,15}$ or $18{:}2^{\Delta 9,12}$ to$18{:}3\ ^{\Delta 9,12,15}$ (Ohlrogge and Browse, 1995). *B. juncea* contained 10 of *ω-3 FAD* genes, *B. rapa* and *B. nigra* both had 6, and 3 were found in *A. thaliana*.

Among 57 *BjuFAD* genes, 54 were unevenly distributed in 16 chromosomes of *B. juncea* except for A06 and B04, and 3 were located in unassembled genomic scaffolds (Supplementary Fig. S1 and Table S2). Chromosome B01 and B07 both contained seven genes, chromosome A08 had 6 genes, and 5 genes were mapped in each of chromosome A03 and 05. Chromosome A01, B02, B05, B06 and B08 all comprised 3 genes, 2 genes were located in each chromosome of A04, A09 and A10, and only 1 gene was mapped in chromosome A02, A07 or B03. A total of 28 duplicated *BjuFAD* gene pairs were found in *B. juncea* genome (Supplementary Table S6). According to chromosome location, 25 repeated pairs belonged to segmental duplications, and another three was tandem duplications.

Previous studies showed the evidence for genome duplication and triplication events in *Brassica* species, which were followed by diploidization that contained substantial genome reshuffling and gene losses (Yang et al. 2016; Chalhoub et al. 2014; Liu et al. 2014; Wang et al. 2011). *B. juncea* is an allotetraploid that is originated from hybridization between the diploids *B. rapa* and *B. nigra* (Yang et al. 2016). In this study, a total of 57, 45 and 46 *FAD* genes were identified in *B. juncea*, *B. rapa* and *B. nigra* genomes, respectively, whereas Arabidopsis contained 25 *FAD* genes. Interesting, in ADS/FAD5 subfamily, *B. juncea* had only 11 members that was less than *B. rapa* (20) and *B. nigra* (19) and similar to Arabidopsis (9). This result needs to be further validated by systemically cloning of *ADS* gene family in *Brassica* species.

Gene structure analysis of *BjuFAD* genes indicated that both exon–intron organizations and intron patterns within BjuFAD subfamily were relatively conserved (Supplementary Fig. S2). For instance, *FAB2*, *ADS*, *FAD2*, *FAD3/7/8*, *FAD4*, *FAD6*, *DES1L* and *SLD* subfamily encompassed 1–5, 2–8, 0–3, 4–7, 0–1, 9–15, 1 and 0 introns.

Twenty conserved motifs were identified in 57 BjuFAD proteins using MEME web server (Supplementary Fig. S3 and Table S7). Thirteen motifs belonged to FA_desaturase_2, FA_desaturase, TMEM189_B_dmain, and Cyt-b5 domains, transmembrane region and an unknown function domain (DUF3474), while other motifs were not annotated. Motif 16, 8, 1, 7 and 15 were harbored in BjuFAB2s, motif 20, 6, 19, and 14 for BjuADSs, motif 10, 4, 8, 2 and 17 for BjuSLDs, motif 3 and 2 for BjuDES1Ls, motif 19, 3 and 5 for BjuFAD4s, motif 11, 3, 12 and 2 for BjuFAD6s, motif 11, 3, 15, 13, 9, 2 and 8 for BjuFAD3/7/8, and motif 11, 3, 18, 9 and 2 for BjuFAD2s.

Prediction of the subcellular locations indicated that all members for FAD4, FAD6, FAD7 and FAD8 subfamily in *B. juncea* were harbored in chloroplast (Supplementary Table S2). Additionally, BjuADS (except for BjuADS5) and BjuFAB2 (except for BjuFAB2.3/4/12) subfamily were also located in chloroplast. Interesting, unlike AtFAD2/3, BjuFAD2.3/4 and BjuFAD3.4/5 were harbored in chloroplast and mitochondria, respectively. These plastid-located FAD proteins all contained 10-82 aa SP in N-termini. No occurrence of SP was detected in the rest of BjuFAD proteins, suggesting they might be located in cytoplasm and other subcellular positions.

Gene structure and MEME motifs of *BjuFAD* genes were highly conservative in each subfamily, which is similar to those in previous reports (Chi et al. 2011; Liu et al. 2015; Dong et al. 2016). In general, the soluble and membrane-bound BjuFADs harbored two and three histidine boxes, respectively, which function in binding the di-iron complex essential for maintaining catalytic activity of enzymes (Shanklin et al. 1994; Shanklin and Cahoon 1998). Unexpectedly, microsomal BjuFAD2.3/4 and BjuFAD3.4/5 were predicted to be located in chloroplast and mitochondria, respectively, which are not consistent with ER location of other plant FAD2/3 (Zhang et al. 2009; Liu et al. 2012; Lee et al. 2016). This necessitates confirming the accurate subcellular location of these four proteins in further research.

cis-Acting regulatory elements function as molecular switches in transcriptional regulation of a variety of biological processes, e.g., developmental process, and responses of abiotic stress and phytohormone (Yamaguchi-Shinozaki and Shinozaki 2005). Therefore, to further investigate the transcriptional regulation and potential functions of *BjuFAD* genes, *cis*-acting regulatory elements in BjuFAD promoters were predicted and analyzed by PlantCARE (Supplementary Table S8 and Fig. S4). Results indicated that *BjuFAD* promoters all contained common *cis*-regulatory elements, e.g. CAAT-box, TATA-box, and light-responsive elements. Numerous cell- and tissue-specific *cis*-elements were identified in upstream regions of *BjuFADs*, including root-, shoot-, endosperm-, and meristem-specific expression elements. BjuFAD promoters also had multiple *cis*-elements involved in plant hormone response, e.g. MeJA-, SA-, ABA-, gibberellin-, auxin-, and ethylene-responsive elements. In addition, BjuFAD promoters harbored three or more *cis*-elements associated with stress responses, such as heat, low-temperature, salt, drought, dehydration, and fungal elicitor response elements. This suggested that there existed a wide diversity for the transcriptional regulation of *BjuFAD* genes in plant growth and development as well as in response to hormones and various stresses.

A total of 16 SSR loci were detected in 57 *BjuFAD* genes/promoters (Supplementary Table S9), which included 5 types, i.e. mono-, di-, tri-, hepta- and octa-nucleotide motifs. Of 16 SSR motifs, 5 were located in intron, only 1 was mapped to exon and other SSR loci were located in promoters. In addition, 4 SSR loci were present in *BjuFAB2* subgroup, 5 for *BjuADS* subgroup, 1 for each of *BjuSLD*, *BjuDES1L* and *BjuFAD2* subgroup, and 2 for either *BjuFAD3/7/8* or *BjuFAD4* subgroup. Substantial data indicated that SSRs in 5′-UTR, 3′-UTR and intron, functioned in regulating gene transcription and translation, and mRNA splicing (Li et al. 2004). Therefore, there is a need to further study their regulation function. This can also provide insights into quality improvement of *B. juncea* seed oil using marker-assisted breeding based on FAD SSR loci.

In conclusion, we analyzed the phylogeny, exon–intron organization, conserved motifs, subcellular locations, SSR motifs, and *cis*-regulatory elements of the promoters in *BjuFAD* gene family at the whole-genome level. This work will provide a molecular basis for quality improvement of *B. juncea* seed oil by metabolic engineering.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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